

SYLLABUS & SCHEME OF EXAMINATION

**BACHELOR IN MEDICAL LABORATORY
TECHNOLOGY (BMLT)**



SESSION 2021-22 & ONWARDS

REGULATION

REGULATION FOR DEGREE COURSE (3 YEARS) IN PARAMEDICAL SUBJECTS

The objective of the courses

1. The training of the candidate registering for three years degree course is aimed to carry out all routine diagnostic / therapeutic tests on modern hospital laboratory
2. The Degree course is of three years
3. Eligibility for admission in Para medical Degree course
 - a) The candidate must have passed 10+2 with at least 45% marks in aggregate of Physics Chemistry and Biology (P C B) or **DMLT** with 50% marks for BMLT will be eligible
4. In case of SC/ST/OBC candidates, 5% marks relaxation will be given for the admission in degree course
5. Number of seats will be 50.
6. The minimum age for admission shall be **17 years** on 31 December of the Academic year of admission
7. The selection of candidate will be on the basis of the entrance examination or in the absence of entrance exam, purely on merit list prepared on the basis of marks obtained in qualifying examination
8. The admissions of students into degree programs should be completed by **31st July**. There shall be **minimum 240 teaching days** in one academic year.
9. **Attendance of students:** The student should have 75% attendance in theory and 75% attendance in practical to be eligible to appear in university examination.
However, only Dean/ Principal of the college are empowered to condone 10% of attendance on valid grounds. Monthly attendance of students has to be sent to the university every month.
10. **Total Course Duration:** A student has to complete the course that he/she has joined with in the double the duration of the actual time taken for the completion of the course i.e. - Six Years for Degree Course.
11. **Medium of instruction and examination:** In case of degree course the medium of instruction will be English and the medium of examination will be English.
12. There will be one Internal and University Examinations in one academic session.
13. **Passing Marks of Examination:**
 - The passing marks of examination would be 50% for each subject and also in total marks obtained for Degree course. The candidate has to pass in theory and practical examination separately. For theory paper the internal assessment marks, viva-voce and theory examination marks will be counted.
 - The candidate has to pass separately in each subject in internal assessment examination (with 50% marks) in order to be eligible to appear in university examinations.
 - Regarding Grace Marks it was decided that total weightage of grace marks would be 5 (five), and that grace marks can be split between the subjects. There will not be any grace marks for the Practical examination.
 - Grace marks will not be added to total marks of the candidate. In Supplementary examination also similar pattern of grace marks will be followed.
14. Re-totaling & Re-valuation (Review) both should be allowed as per the university rules

15. Other rules regarding conduct of examination will be as per the university

REGULATION/notifications

16. Permission will be given to start the courses after the permission from M P Paramedical council (Department of Medical education, Government of M. P.)

17. Syllabus and scheme of examination shall be decided by the board of studies and Academic council of the university from time to time as per guidelines from M P Paramedical council

18. In order to be an examiner a faculty member should have minimum three years of teaching experience and not below the rank of the Associate Professor/ Assistant Professor.

19. The examination papers (Theory) will be evaluated centrally at the university

20. The examiner who evaluates the theory copy preferably be appointed as the external for practical examinations for the said course.

21. Degree students will be eligible for supplementary examination.



FACULTY OF PARAMEDICAL SCIENCE

BHABHA UNIVERSITY, BHOPAL

(Approved by M.P. Paramedical Council)

NH-12, Hoshangabad Road, Bhopal

<https://www.bhabhauniversity.edu.in/>

email: bhabhauniversitybpl@gmail.com

SYLLBUS & SCHEME OF EXAMINATION FOR BMLT 1ST YEAR

		Internal Assessment	University Examination	
Sub. Code	Subjects	Theory	Theory & Practical With Viva	Total
BMLT101	Human Anatomy & Physiology	100 Marks	200 Marks	300
BMLT102	General Biochemistry	100 Marks	200 Marks	300
BMLT103	General Pathology, Hematology & Clinical Pathology	100 Marks	200 Marks	300
BMLT104	General Microbiology, Immunology & Parasitology	100 Marks	200 Marks	300
Total				1200

SYLLBUS & SCHEME OF EXAMINATION FOR BMLT 2nd YEAR

		Internal Assessment	University Examination	
Sub. Code	Subjects	Theory	Theory & Practical With Viva	Total
BMLT201	Clinical Biochemistry	100 Marks	200 Marks	300
BMLT202	Immunohematology & Blood Banking	100 Marks	200 Marks	300
BMLT203	Histopathology & Cytopathology	100 Marks	200 Marks	300
BMLT204	Bacteriology, Virology & Mycology	100 Marks	200 Marks	300
Total				1200

SYLLBUS & SCHEME OF EXAMINATION FOR BMLT 3rd YEAR

		Internal Assessment	University Examination	
Sub. Code	Subjects	Theory	Theory & Practical With Viva	Total
BMLT301	Advance Biochemistry & Molecular Pathology	100 Marks	200 Marks	300
BMLT302	Advance Histopathology & Cytopathology	100 Marks	200 Marks	300
BMLT303	Clinical Hematology	100 Marks	200 Marks	300
BMLT304	Clinical& Applied Microbiology	100 Marks	200 Marks	300
BMLT305	Dissertation	50 Marks	--	50
Total				1250

NOTE: Internal Assessment marks will be added in theory marks; candidate have to get min. 50% marks i.e.- 100 marks in theory and internal assessment collectively for passing the examination and in practical he/she should get 50% marks i.e.-50 marks to get pass.

All theory papers will of maximum 100 marks and time duration 3Hrs.

Pattern of University Theory Examination will be as under;-

Type of Questions	Number of Questions	Marks for each Question	Total Marks
Very Short Questions	10	2	20
Short Questions	5	10	50
Long Question	2	15	30
			100

SYLLABUS & SCHEME OF EXAMINATION FOR BMLT 1ST YEAR

S. No.	Subject Code	Subject	Theory				Practical		Total Marks
			UE	IA	Total	Minimum Marks	Practical & Viva Voce	Minimum Marks	
1	BMLT101	Human Anatomy & Physiology	100	100	200	100	100	50	300
2	BMLT102	General Biochemistry	100	100	200	100	100	50	300
3	BMLT103	General Pathology, Hematology & Clinical Pathology	100	100	200	100	100	50	300
4	BMLT104	General Microbiology, Immunology & Parasitology	100	100	200	100	100	50	300
Grand Total									1200

Paper: I Human Anatomy & Physiology

Course Code: BMLT101

Course Contents: Theory

An Introduction to the Human Body: Definition and divisions of anatomy. Anatomical terms related to positions, planes and movements.

Cellular & Tissue Level of the Organization: Cell- Definition, Structure, Function and cell division. Tissue- Definition, classification, structure and function of Epithelial Tissue, Connective Tissue, Muscular Tissue, Nervous Tissue and Glands.

Skeletal System: Bone- Definition, structure and function of bones. Axial and appendicular skeleton. Joints- Definition, classification and function of joints.

Muscular System: Definition, classification, structure and function of muscles.

Digestive System: Structure and function of esophagus, stomach, small and large intestine, liver pancreas. Structure and function of accessory organ of digestive system. Composition and function of saliva, gastric juice, bile juice and pancreatic juice. Bilirubin and Jaundice. Digestion and absorption of carbohydrate, protein and lipids.

Renal System: Structure and function of Kidneys, ureters, urinary bladder and urethra. Nephron- structure and function. Mechanism of urine formation.

Respiratory System: Structure and function of Nose, Pharynx, larynx, trachea, lungs, . Mechanism of respiration. Exchange of gases. Pulmonary function test.

Circulatory: Heart- structure and function. Heart rate, cardiac cycle, blood pressure, pulse rate, etc. Pulmonary and systemic circulation. Blood vessels- Aorta, artery, vein, etc. difference between artery and vein. Blood and Blood group system.

Lymphatic System: Lymph node, spleen and Reticuloendothelial system.

Nervous System: Structure and function of Central nervous system- Brain and spinal cord, Peripheral nervous system and autonomic nervous system. Neuron- structure and function.

Endocrine System: Structure, functions and hormones of pituitary, thyroid, parathyroid, pineal, thymus, pancreatic, adrenal gland, etc.

Special Sense: Structure and functions of skin, eye, ear, nose and tongue.

Reproductive System: Male reproductive organs- Structure and function of testis, Duct system, prostate and urethra. Spermatogenesis, semen. **Female reproductive organs-** Structure and function of Uterus, ovary vagina, cervix. Menstrual cycle, Oogenesis, Fertilization. Mammary gland- Structure and function. Fertility and Infertility. Fertility Control

Course Contents: Practical

Determination of ABO and Rh grouping.

Determination of body temperature, pulse rate and blood pressure.

Recommended Books

B.D. Chaurasia's General Anatomy- Krishna Garg

Ross & Wilson Anatomy & Physiology- Waugh Grant

Applied Anatomy & Physiology for Nursing Students- Krishna Garg

Essential of Medical Physiology- K. Sembulingam

Paper- II General Biochemistry

Course Code: BMLT102

Course Contents: Theory

Introduction to Medical Laboratory Technology. Ethics, responsibilities of Medical Laboratory Technologists. Safety measures and first aid. Units of measurement. S.I Units
Laboratory Equipments and glassware- cleaning, care and maintenance.
Basic instrument- Colorimeter, Spectrophotometer.
Collection of biological specimens and separation of serum and plasma.
Reagents and Solutions: Normal solution, Molar solution Percent solution, buffer solution, Indicators, distilled water, etc- their preparation and storage.
Carbohydrate- Definition, classification, structure, properties and their function of Monosaccharides, disaccharides and polysaccharides.
Amino acid, Protein and Plasma protein- Definition, properties, classification and function.
Lipid- Definition classification, structure, properties and function of lipids.
Enzyme- Definition, classification, properties and function of enzymes.
Vitamins and Minerals- Water soluble and fat soluble vitamins.

Course Contents: Practical

Blood Sugar, Total protein, Albumin, Cholesterol, Triglyceride,
Uric acid, Urea, Creatinine, Bilirubin, SGPT, SGOT, etc

Recommended Books

Textbook of Medical Biochemistry for Medical Students- DM Vasudevan
Biochemistry- Satyanarayana
Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)
Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)
Tietz Fundamental of Clinical Chemistry and Molecular Diagnostics- Nader Rifai
Varley's Practical Clinical Biochemistry- Harold Varley

Paper- III General Pathology, Hematology & Clinical Pathology

Course Code: BMLT103

Course Contents: Theory

General Pathology

Introduction and definition

Cell injury, cell death and cellular adaptation

Inflammation: Definition, Acute and chronic inflammation

Healing and Repair: Definition, different phases of healing.

Hemodynamic Disorders: Oedema, hyperemia, thrombosis, ischaemia & infarction

Neoplasm: Definition, difference between benign and malignant tumor.

Hematology

Introduction to hematology.

Anticoagulants, Diluting fluids, Romanowsky stain and other reagents.

Collection and preservation of blood.

Basic instruments- Microscope, Hemoglobinometer, Hemocytometer,

RBC pipette, WBC pipette, Hb pipette, ESR pipette and tube, etc.

Blood- Definition, properties, composition and function.

Haematopoiesis- Erythropoiesis, Leucopoiesis and Thrombopoiesis

Hemoglobin-Definition, Structure, synthesis, function and its type.

Polycythemia.

Morphology of blood cells- Normal and abnormal.

Qualitative and Quantitative disorders of Leukocytes

Bone marrow examination- Collection, composition, functions.

Haemoparasites.

Normal and absolute values in hematology.

Haemostasis and thrombosis. Blood coagulation factor and mechanism of clotting.

Quality assurance in hematology.

Clinical Pathology

Introduction- Instruments and reagents used in Clinical Pathology

Urinalysis and Stool Examination- Physical, chemical and microscopic examination.

Cerebrospinal Fluid, Semen and other body fluids examination.

Course Contents: Practical

Microscope- Part, function and care.

Total Erythrocyte count, Hemoglobin estimation, PCV, Red cell indices, ESR, Total Leukocyte count, Total Platelet count, Reticulocyte count, Absolute Eosinophil count, BT, CT.

Differential leukocyte count

Peripheral blood smears: Thin and Thick smear, wet smear and buffy coat smear preparation for parasitic detection. Staining method by Romanowsky stain.

Bone marrow smears Preparation and staining methods

Physical, chemical and microscopic examination of urine, stool, etc

Recommended Books

Textbook of Pathology – Harsh Mohan

Pathology Practical- Harsh Mohan

Pathology Practical- P. Chakraborty

Text and Practical Hematology for MBBS- Tejindar Singh

Hematology for Students and Practitioners- Ramnik Sood

Essentials of Clinical Pathology- S. M. Kawthalkar

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Paper- IV General Microbiology, Immunology & Parasitology

Course Code: BMLT104

Course Contents: Theory

General Microbiology

Historical Introduction: Contribution of some important Scientists.

Laboratory instruments- their used, care and maintenance.

Microscopy and Staining method. Culture Media and Culture Method

Morphology of Bacteria. Nutrition and growth of bacteria.

Identification of bacteria. Antimicrobial susceptibility testing.

Microbial Pathogenicity and Epidemiology of infectious disease

Sterilization, Disinfection and Antiseptic.

Hand Hygiene. Biomedical waste management.

Collection and transportation of microbiological specimen

General properties, classification and laboratory diagnosis of Virus, Fungi and Bacteria.

Immunology

Immunity. Antigen and Antibody (Immunoglobulin)

Antigen-antibody reaction. Structure and function of immune system.

Complement system, Immune response

Hypersensitivity, Autoimmunity, Immunodeficiency disease, Immunoprophylaxis

Parasitology

Introduction- General properties, classification and laboratory diagnosis of parasites.

Protozoology: Amoeba (Entamoeba, etc), Flagellates (Giardia, Trichomonas, Leishmania, Trypanosoma, etc), Sporozoa (Plasmodium, Toxoplasma, etc), Ciliate (Balantidium).

Helminthology: Cestodes (Taenia, Hymenolepis, Echinococcus, etc), Trematodes (Fasciola, Schistosoma, etc) and Nematodes (Ascaris, Trichuris, Ancylostoma, Strongyloides, Enterobius, Wuchereria, etc)

Course Contents: Practical

Autoclave and Hot air oven- Structure and sterilization process.

Gram stain, ZN Stain and Albert stain. Hanging drop preparation for motility of bacteria.

ASO, CRP, VDRL, Widal, RA Factor, Pregnancy test, other serological test.

Slide preparation for stool microscopy and identification parasitic ova and eggs.

Recommended Books

Complete Microbiology for MBBS- C.P. Baveja

Practical Microbiology- D.R Arora

Ananthanarayan and Paniker's Textbook of Microbiology- Reba Kanungo

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Bailey & Scott's Diagnostic Microbiology- Patricia M. Tille

SYLLABUS & SCHEME OF EXAMINATION FOR BMLT 2ND YEAR

S. No.	Subject Code	Subject	Theory				Practical		Total Marks
			UE	IA	Total	Minimum Marks	Practical & Viva Voce	Minimum Marks	
1	BMLT201	Clinical Biochemistry	100	100	200	100	100	50	300
2	BMLT202	Immunohematology & Blood Banking	100	100	200	100	100	50	300
3	BMLT203	Histopathology & Cytopathology	100	100	200	100	100	50	300
4	BMLT204	Bacteriology, Virology & Mycology	100	100	200	100	100	50	300
Grand Total									1200

Paper- I Clinical Biochemistry

Course Code: BMLT201

Course Contents: Theory

Water and Mineral metabolism. Acid- Base balance.

Hormones- Growth hormone, Prolactin, Follicle stimulating hormone, Luteinizing hormone, Thyroid stimulating hormone, Oxytocin, Estrogen, Progesterone, etc.

Liver function test, Renal function test, Cardiac profile, Thyroid profile, Diabetic profile, Pancreatic function test, Gastric function test, etc

Flame-Photometer- Structure, function, care and maintenance.

Quality control in Clinical Biochemistry.

Course Contents: Practical

Glucose tolerance test, Urea and Creatinine clearance test,

Estimation of Amylase, Lipase, Acid & Alkaline Phosphate, GTT, etc

Estimation of Sodium, Potassium, Chloride, Calcium, HbA_{1C}.

Estimation of T₃, T₄, TSH, etc

Recommended Books

Textbook of Medical Biochemistry for Medical Students- DM Vasudevan

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Tietz Textbook of Clinical Chemistry and Molecular Diagnostics- Nader Rifai

Tietz Fundamental of Clinical Chemistry and Molecular Diagnostics- Nader Rifai

Paper- II Immunohematology & Blood Banking

Course Code: BMLT202

Course Contents: Theory

Historical Introduction to Blood bank.

Anticoagulants and Instruments used in blood bank.

Blood collection, processing and storage.

Blood Group System- ABO System and Rh System. Other minor blood groups

Blood grouping techniques- Forward and reverse grouping.

Blood donation- Blood donor and their types.

Criteria for selection and rejection of donor.

Testing of blood donor: Coomb's test, Screening tests for transfusion transmitted disease, etc

Compatibility testing- Cross matching

Hemolytic disease of newborn

Blood transfusion reaction- Types and their investigation.

Blood components and their preparation: Whole blood, Blood Components and Blood Derivatives.

Aphaeresis

Course Contents: Practical

ABO blood grouping and Rh typing. Preparation of red cell suspension.

Coomb's Test, Cross Match, D^u test

Transfusion transmitted tests- HBsAg, HIV, HCV, etc

Recommended Books

Essential of Blood Banking and Transfusion Medicine- Ganga S. Pilli

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Modern Blood Banking and Transfusion Medicine- Denise M. Harmening

Paper- III Histopathology & Cytopathology

Course Code: BMLT203

Course Contents: Theory

Histopathology

Introduction and Examination of tissue: Reception, recording, labeling and grossing
Basic instrument: Microscope, Tissue processor, Wax bath, Water bath, Microtome, etc.
Microtome knife and knife sharpening.
Fixatives and Fixation of tissue
Decalcifying fluids and Decalcification of tissue.
Tissue processing.
Embedding media and Embedding.
Microtomy and section cutting.
Section cutting faults and their cause and remedies.
Slide preparation and Staining procedure and Mounting

Cytopathology

Introduction to Exfoliative cytology especially for gynecological specimen.
Collection, slide preparation and staining procedure in cytology.

Course Contents: Practical

Decalcification, Fixation process.
Tissue processing, Microtomy.
Sharpening of knife.
H & E staining, Pap staining, etc

Recommended Books

Manual of Histological Techniques- Santosh Kumar Mandal
Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)
Bancroft's Theory and Practice of Histological Techniques- S. Kim Suvarna
Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Paper- IV Bacteriology, Virology & Mycology

Course Code: BMLT204

Course Contents: Theory

Bacteriology

Gram-positive cocci (Staphylococcus, Streptococcus, Pneumococcus, etc),

Gram-negative cocci (Neisseria, etc),

Gram-positive bacilli (Corynebacterium, Mycobacterium, Clostridium, Anaerobes, Spirochaetes, etc),

Gram-negative bacilli (Escherichia, Klebsiella, Proteus, Shigella, Salmonella, Vibrio, Pseudomonas, Haemophilus, etc),

Other group of bacteria (Spirochetes, Chlamydia, etc).

Virology

DNA viruses- Poxviruses, Herpesviruses, Adenoviruses, Parvoviruses, Papovaviruses,

RNA viruses- Picornaviruses, Orthomyxoviruses, Paramyxoviruses, Arboviruses, Rhabdoviruses, Human immunodeficiency virus, Coronaviruses, Filoviruses, Rubivirus, etc

Miscellaneous viruses- Hepatitis virus, Oncogenic virus.

Mycology

Superficial mycoses (Dermatophytosis, Tinea, etc),

Subcutaneous mycoses (Mycetoma, etc),

Systemic mycoses (Histoplasmosis, etc),

Opportunistic mycoses (Candida, Cryptococcosis, Aspergillosis, etc)

Course Contents: Practical

Preparation of different culture media. Antimicrobial sensitivity test

Identification of bacteria by biochemical tests.

Immuno-Serological tests for detection of virus.

India ink preparation, KOH preparation, Lactophenol cotton blue staining, etc

Recommended Books

Complete Microbiology for MBBS- C.P. Baveja

Ananthanarayan and Paniker's Textbook of Microbiology- Reba Kanungo

Bailey & Scott's Diagnostic Microbiology- Patricia M. Tille

SYLLABUS & SCHEME OF EXAMINATION FOR BMLT 3RD YEAR

S. No.	Subject Code	Subject	Theory				Practical		Total Marks
			UE	IA	Total	Minimum Marks	Practical & Viva Voce	Minimum Marks	
1	BMLT301	Advance Biochemistry & Molecular Pathology	100	100	200	100	100	50	300
2	BMLT302	Advance Histopathology & Cytopathology	100	100	200	100	100	50	300
3	BMLT303	Clinical Hematology	100	100	200	100	100	50	300
4	BMLT304	Clinical & Applied Microbiology	100	100	200	100	100	50	300
5	BMLT305	Dissertation	--	50	--	--	--	--	50
Grand Total									1250

Paper- I Advance Biochemistry & Molecular Pathology

Course Code: BMLT301

Course Contents: Theory

Cancer and Tumor marker. Analysis of calculi.

Nucleic acids- Structure and function of DNA and RNA. Replication of DNA.

Cytogenetic- Karyotyping

Inborn Error of Metabolism

Polymerase chain reaction,

Radioimmunoassay, ELISA,

Electrophoresis, Chromatography

Quality control and Automation in clinical biochemistry.

Course Contents: Practical

Alpha-fetoprotein, Prostate specific antigen,

Carcinoembryonic antigen, CA 125, etc.

Recommended Books

Textbook of Medical Biochemistry for Medical Students- DM Vasudevan

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Tietz Fundamental of Clinical Chemistry and Molecular Diagnostics- Nader Rifai

Medical Genetics- G.P. Pal

Paper- II Advance Histopathology & Cytopathology

Course Code: BMLT302

Course Contents: Theory

Fine needle aspiration cytology

Frozen Section and Cryostat

Staining for microorganisms.

Staining for Carbohydrate, Pigments, minerals, Amyloid, Lipids, etc

Cytological study from discharges, scrapping and aspirates.

Autopsy and Museum Technique

Automation in histopathology

Course Contents: Practical

PAS Stain, Shorr's stain, etc

Staining in Exfoliative gynecological cytology

Recommended Books

Manual of Histological Techniques- Santosh Kumar Mandal

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)

Bancroft's Theory and Practice of Histological Techniques- S. Kim Suvarna

Paper- III Clinical Hematology

Course Code: BMLT303

Course Contents: Theory

Anemia- Definition, classification. Iron Deficiency Anemia, Megaloblastic anemia, Hemolytic Anemia, etc and their laboratory diagnosis. Paroxysmal nocturnal hemoglobinuria.
Thalassemia- Clinical feature and laboratory diagnosis.
Leukemia- Acute Myeloid Leukaemia, Chronic Myeloid Leukaemia, Acute Lymphoid Leukaemia and Chronic Lymphocytic Leukaemia.
Plasma cell neoplasms, etc
Normal Haemostasis. Platelet disorders and Platelet function test.
Coagulation and Bleeding disorders and their laboratory diagnosis.
Automation in hematology.

Course Contents: Practical

LE Cell Phenomenon, Fetal Hemoglobin, G6PD, Heinz bodies, Sickling test, Osmotic fragility test, PAS Stain, HAM test NESTROF test, etc
PT, PTT, APTT,

Recommended Books

Text and Practical Hematology for MBBS- Tejindar Singh
Hematology for Students and Practitioners- Ramnik Sood
De Gruchy's Clinical Haematology in Medical Practice- Saxena, Pati, Mahapatra
Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)
Medical Laboratory Technology Methods and Interpretation- Ramnik Sood (2 Vols)
Wintrobe's Clinical Hematology- John P. Greer

Paper- IV Clinical & Applied Microbiology

Course Code: BMLT304

Course Contents: Theory

Clinical microbiology

Normal microbial flora.

Musculoskeletal infection,

Blood and Cardiovascular infection,

Gastrointestinal and Hepatobiliary infection,

Respiratory tract infection,

Central nervous system infection,

Genitourinary tract infection and sexual transmitted disease.

Opportunistic Infections. Hospital acquired infection- Definition and types.

Bacteriology of Water, Milk, Food and Air.

Ethics and confidentiality. Quality control in microbiology.

Automation in microbiology.

Zoonoses. Vehicles and Vector.

Experimental animals- Care and handling.

Diagnostic Microbiology

Course Contents: Practical

Examination of Sputum, Urogenital, throat, pus, etc

Recommended Books

Complete Microbiology for MBBS- C.P. Baveja

Ananthanarayan and Paniker's Textbook of Microbiology- Reba Kanungo

Textbook of Medical Laboratory Technology- P.B. Godkar- (2 Vols)

Bailey & Scott's Diagnostic Microbiology- Patricia M. Tille

Dissertation

- Microscopes
- Analytical Instruments
- Sterilization Instruments
- Hematological Instrument
- Automation
- Centrifuge Machine
- Microtome
- Water Bath
- Incubator
- Pipettes and Micropipettes
- Chromatography
- Electrophoresis
- Tissue Processor
- PCR
- ELISA
- Biomedical Waste Management

Note: There will be Internal Assessment of following Dissertation of maximum 50 marks and the dissertation marks have to sent to university prior to theory examination.